

Question:

John decides to build a program capable of identifying cancer cells. He gets initial funding of \$100,000 for the first 5 months of research. His research ends up taking 10 times that long and every month after those first 5 took 50% more funding per month. How much did his research cost?

Reasoning + Answer:

<think>John gets initial funding of \$100,000 for the first 5 months of research. Since his research ends up taking 10 times that long, the number of months his research takes is: $\lfloor 5 \times 10 = 50 \text{ months} \rfloor$ The first 5 months of research cost \$100,000, so the number of months after the first 5 months is: $\lfloor 50 - 5 = 45 \text{ months} \rfloor$ For the months after the first 5, every month took 50% more funding per month, which means the funding per month for these months is: $\lfloor 100,000 \times 1.5 = 150,000 \text{ dollars per month} \rfloor$ The cost for the 45 months after the first 5 months is: $\lfloor 45 \times 150,000 = 6,750,000 \text{ dollars} \rfloor$ Thus, the total cost of his research is the sum of the initial funding and the funding for the subsequent months: $\lfloor 100,000 + 6,750,000 = 6,850,000 \text{ dollars} \rfloor$ </think><answer>6850000</answer>

